Decomposition and synthesis

There are many different types of chemical reactions. **Decomposition** and **synthesis** are just two. 'Decompose' means to break down, and 'synthesise' means to combine.

Reaction	Definition	Equation
decomposition reaction	A reaction where one compound breaks down into two or more simpler substances	AB ⟶ A + B
synthesis reaction	A reaction where two or more substances combine to form a more complex substance	A + B ⟶ AB

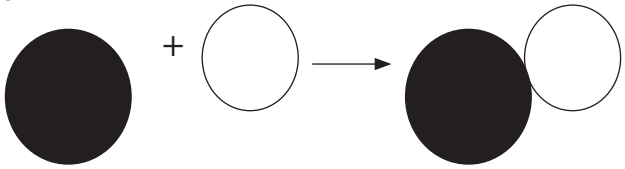
1 Use the words in the word bank to complete the following sentences.

- Decomposition reactions require e_____ , either in the form of h_____ , light or e_____.
- The a_____ energy causes the reacting substance to b_____ down.
- Synthesis reactions are e_____.
- C_____ decompose but e_____ cannot.

Word bank
absorbed
compounds
energy
elements
heat
break
endothermic
electricity

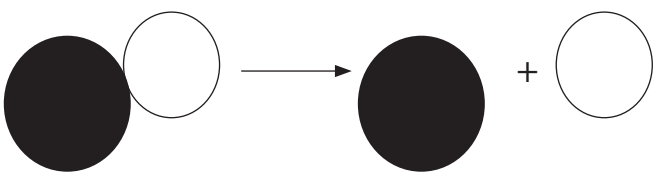
2 Which type of reaction does each diagram represent?

a



_____

b



_____

3 Below are some different word equations. Decide whether each shows a synthesis or decomposition reaction.

a magnesium + oxygen $\xrightarrow{\text{heat}}$ magnesium oxide _____

b calcium carbonate $\xrightarrow{\text{electricity}}$ calcium oxide + carbon dioxide _____

c water ⟶ hydrogen gas + oxygen gas _____

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